THE CROSS-STRAIT BRAIN DRAIN: THE RESILIENCY OF THE TAIWANESE ACADEMIA-INDUSTRIAL COMPLEX AGAINST GEOPOLITICAL RISK

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Gov 1982: Chinese Foreign Policy

Term Paper

December 19, 2023

Introduction

According to the Economist, between 2014 and 2019, “over 3,000 semiconductor workers–7% of Taiwan’s chip workforce, moved to China.” As China invests unprecedented amounts of capital into its semiconductor industry, the central government has come to find it still faces the greatest bottleneck in human capital development. Even with the billions in liquidity and low-rate loans, the Chinese industrial policy still struggles to move up the vertical supply chain towards the highest profit-margin fabs. To precisely complete the hundreds of steps involved in cutting-edge wafer fabrication, China still needs to instigate mass skill-based tech transfer. As such, Beijing scours Hsinchu Science Park for new recruits, offering incredible financial compensation in exchange for semiconductor know-how.

In response to this cross-strait brain drain, the Taiwanese government has cracked down on high-skilled migration flows. In early 2022, the Taiwanese Executive Yuan approved amendments to the National Security Act that criminalized “economic espionage” and the infringement of “core technologies” to China and other “foreign enemy forces.” These new amendments would require individuals and organizations of interest to gain government approval before visiting China, along with punishment guidelines that could sentence offenders with up to 12 years in jail.

However, while wage incentives and allegiances to the Taiwanese semiconductor industry have been historically cited as push-and-pull drivers of high-skilled migration between Taiwan and China, I argue it is largely geopolitical risk that acts as the most significant explanatory variable of these high-skilled migration patterns.

In the following sections, I will discuss the previous works and rationale behind my methodology, along with the theoretical framework I use for hypothesis testing. Then, I will analyze the interplaying changes of my proposed independent variables against the changes of cross-strait talent migration. To do so, I give context on the last two decades of Chinese efforts towards developing its human capital resources, discuss the increasing relative success of such efforts in recruiting foreign talent leading up to 2019, and how these efforts were upended by other stronger explanatory factors in the post-2019 period.

Literature Review and Methodology

Despite the strongarm legislation being passed, the Executive Yuan lacks information on the migrant demographic and precise motivations for migration. Previous literature has framed migration around theoretical models citing economic, social, and institutional factors (Wang 1186). Wage and compensation-based incentives have historically been considered the primary driver of migration leveraged by Chinese firms. Chinese roles are known to offer salaries five, six times the industry standard, and the Taiwanese government has loudly cracked down on firms for engaging in illegal recruiting practices. In 2021, headlines broke for the crackdown of Beijing-based cryptocurrency startup Bitmain Technologies Ltd., which was suspected for having illegally recruited hundreds of Taiwanese engineers over three years (Wu). Still, it remains unclear whether high-skilled talent have increasingly chosen to leave China and work in Taiwan due to insufficient financial compensation, fear of social pressure surrounding accusations of espionage, uncertainty in the geopolitical environment of China, or yet other reasons.

In my analysis, I will compare changes in various independent variables against the post-2019 decrease in high-skilled talent working in China. Given the recent nature of this topic, this split of the pre- and post-2019 period is not based on previous literature, but instead a logical observation of various global shifts in that year. The post-2019 period saw a significant escalation in geopolitical tensions, heightened by the U.S.-China trade war and Taiwan's wariness of Chinese cyber attacks, leading to increased scrutiny and risks associated with cross-strait talent migration. The Covid-19 pandemic exacerbated these risks, with political and national security concerns significantly influencing talent mobility decisions, potentially overshadowing financial incentives.

Most of my evidence is collected from a variety of primary and secondary sources, with credit given in the Works Cited section. In addition to these third-party sources, I also draw from two original sources of data. First, I conducted interviews with about a dozen critical stakeholders in the Taiwanese semiconductor industry during January 2023 through a grant from the Harvard University Asia Center. Most of these interviews were conducted in Mandarin–I have attached raw interview notes in the Appendix. Second, I created a web scraper for the 104 Job Bank, Taiwan's main job board site. This Python-based web scraper uses Selenium and Beautifulsoup to grab job listings under customizable search queries, and outputs a CSV file with columns describing details of each listing, including location, salary, required skills, role type, and more. The scraper code is publicly available at https://github.com/JessSanChen/104scraper. Unfortunately, I could not find comparable sources of data on Chinese job listings, so I was not able to heavily utilize data collected from this scraper in my analysis.

Theoretical Framework

Dependent Variable: Volume of high-skilled migration between Taiwan and China.

One of the major limitations of this paper is the lack of post-2019 data on the volume of migration and workforce data for high-skilled Taiwanese talent working for Chinese firms. The last major statistic states that, by 2019, there were an estimated 3,000 Taiwanese semiconductor engineers working in China, making up almost 10% of the industry's workforce. While there have not been more updated metrics on this workforce on publicly available sites, we do know that a 2022 ban by the U.S. government will force at least 200 Chinese and Taiwanese engineers to either leave China or give up their U.S. citizenship (Perlez *et al*). Many Taiwanese engineers are reported to be returning to the island on their own volition, and tightened review processes by the Executive Yuan have made it harder for engineers to leave Taiwan for China. Therefore, this paper makes the large, but perhaps reasonable, assumption that since 2019, there has been a significant decrease in Taiwanese talent working in Chinese enterprises.

Independent Variable: Geopolitical risk surrounding the Chinese semiconductor industry.

Geopolitical risk in this context refers to the political and policy-related factors that impact talent mobility, including cross-strait relations, trade policies, national security concerns, and regional political stability. This risk affects perceptions of long-term career sustainability and personal safety, influencing migration decisions.

Geopolitical risk and uncertainty stems from how sovereign governments interact on political and economic fronts. While geopolitical risk has always been present in cross-strait relations, we see that the Chinese environment in the pre-2019 was generally considered to be stable and on a strong growth trajectory. While foreigners and Taiwanese workers were wary of the communist regime, they were more likely to integrate into the more culturally open metropolitans through foreign-firm business ventures or as part of the National Innovation-Driven Development Strategy. In contrast, the post-2019 Chinese environment saw not just Covid-related shutdowns and economic instability, but also a barrage of national security regulations imposed by the Taiwanese government. The U.S. also intervened, with both the Trump and Biden administrations tightening export controls banning key machinery and software tools from entering China.

In this paper, the geopolitical risk variable also encompasses the long-term economic considerations tied to uncertainty and instability. For example, whether an individual or firm feels confident in their five- to ten-year economic prospects if they were to stay in the currently uncertain Chinese environment falls under the geopolitical risk variable. An engineer's fear of losing access to advanced machinery due to U.S. export controls is also a symptom of geopolitical risk.

Alternative Independent Variable: Short-term financial considerations.

Short-term financial considerations refer to salaries, signing bonuses, and other cash-like forms of compensation. In this paper, we mostly refer to monthly and annual salaries, along with annual bonuses, as metrics of short-term financial considerations. The Chinese government also funnels significant amounts of money into providing grants to researchers, paying for flights back to Taiwan, or providing tax exemption benefits–while these factors are discussed, this paper does not weigh them heavily into the overall analysis. Other structural benefits provided by governments, such as relaxed residency attainment requirements, housing subsidies, medical insurance, funds for children's education, and the like are similarly discussed but weighed less heavily in our analysis.

Alternative Independent Variable: The Taiwanese social identity.

In this paper, we define identity in alignment with Abdelal et al's discussion of "Identity as a Variable." As such, identity can be broken down into four interrelated components: content, strength, valence, and salience. The constructivist approach towards identity emphasizes the importance of normative appropriateness in a group–as long as a person acts in accordance with what is to be Taiwanese traits, they are accepted as Taiwanese. The Taiwanese identity is complex and has undergone significant evolution over the past few decades, and is thus a good candidate to show how a constructivist approach helps define a dynamic concept.

The content of Taiwanese identity centers around its history of colonization periods, mix of indigenous, Hokkien, Hakka, and mainland Chinese culture, rapid social development field by democratization, and rapid economic development fueled by its technology industry. The strength of the Taiwanese identity may vary across individuals, but often stem from its democratization, relatively liberal values, and general contrast from the mainland Chinese identity. The valence of Taiwanese identity tends to be positive, given the importance of shared resilience in an environment of diplomatic isolation. Finally, the salience of Taiwanese identity is particularly relevant, as it is often tied to cross-strait relations and questions such as whether to pursue reunification or independence from China.

In this paper, we particularly consider the social identity elements of increasing polarization between Taiwanese versus Chinese identities. Over several administrations, the Taiwanese people have developed a stronger sense of de-Sinification and feeling more protective of a unique Taiwanese identity. Furthermore, an element of Taiwanese identity may also be the shared social purpose, which describes the group's wish to maximize the role of its semiconductor industry on a global stage.

Analysis of the Pre-2019 Period: Relativity mutualistic relations in cross-strait innovation.

1. Two decades of Chinese efforts to foster its domestic supply of high-skilled human capital was met with mixed success, with particular limitations in attracting non-Taiwanese foreigners to relocate to China.

Since the turn of the millennium, Beijing has long identified its lack of high-skilled human capital as a major bottleneck in its national development. As it rapidly developed in industry and in technology over the 1990s, the central government recognized the need to move its economy from being run on cheap labor-based exports toward more knowledge-based industries. Xi Jinping called talent "the first resource" in China's push for "independent innovation" (Zwetsloot).

This drive towards fostering a vibrant talent economy on Chinese soil manifested in a three-prong strategy: fostering domestic talent, attracting overseas Chinese talent to return, and attracting foreign talent to relocate to China. First, the government aimed to foster domestic talent. The rapid advancement of the Chinese education standard had given rise to incredible academic results in metropolitan areas, but had also created huge disparities in access to resources. As such, the Ministry of Education has launched numerous initiatives to push science and engineering degree rates beyond that of their Western counterparts–from 2000 to 2015, the number of STEM degrees granted by Chinese universities over-quadrupled from 360,000 degrees to 1.7 million degrees. Ph.D. recipients also increased rapidly, from 1,022 in 2001 to 2,824 in 2014 (Zwetsloot). Compared to the Western bloc's projected 30% increase in graduates by 2030, the OECD education director predicted China to see a 300% increase in the same time period (Zwetsloot).

That being said, this breakneck acceleration towards awarding degrees has not necessarily translated to greater innovation output. The vice president for research and industrial relationship at China's Tsinghua University warned that the problem was not the quantity, but the quality of education (Leng). In its more advanced degrees, Chinese training programs still fell short in the technical know-how and interdisciplinary background to effectively prepare students for complex industry processes at the frontier of innovation.

As such, much of Beijing's talent strategy has instead centered around the latter two of the three prongs: attracting overseas Chinese and foreigners to relocate to Chinese soil. After the turn of the millennium, the government began to assume a much more active role in talent recruitment. Ministries were reorganized to push this mission to the forefront of the CCP policy agenda. The influential Organization Department was put at the head of this policy effort, and Xi named the goal of attracting talent back to the mainland as one of the three main focus areas of the United Front (Zwetsloot).

Most of the headlining policy initiatives to drive overseas talent onto Chinese soil centered around financial incentives. Under the Thousand Talents Plan (TTP), returning talent receive signing bonuses, high salaries and funding, and other forms of compensation such as financial support for housing and children's education (Zwetsloot). Overseas Chinese companies also receive support from the CCP to further promote and spread awareness of such incentive programs to those abroad. Major cities revised their *hukou* resident permit systems to better accommodate young returning talent relocating to new hubs (Zwetsloot). Over the years, the Chinese ministries have reported success in these efforts–some figures claim 7,000 persons were recruited over 10 years. The Ministry of Human Resources and Social Security claimed successful recruitment of 50,000 returnees by 2017, and the Ministry of Education claimed that 500,000 of 650,000 students studying abroad returned by 2018–though it is unclear how these figures were calculated (Zwetsloot).

However, foreigners were generally less receptive to the same financial incentives offered by the Chinese R&D industry. Under this attitude of "Have Talent, Will Thrive", over a trillion dollars were funneled into the many programs tied under the overarching National Innovation-Driven Development Strategy (Glaser). This national strategy aimed to transform mainland China into an "important gathering place for global high-end talent" by 2050. It offered many of the same financial incentives to foreigners, while also removing many barriers to entry when it came to migration. For example, in 2017, the foreign talents ("R") work visa expanded beyond "urgently needed talents" to include "scientists, entrepreneurs, and leading experts in science and technology industries" (Zwetsloot). Similar to the *hukou* revisions for returning Chinese students, foreigners also received accommodations for relaxed residency requirements. Further financial incentives include preferential individual income tax policies, which allowed many of the foreigners' financial benefits to become tax-exempt (Zhou & Huang). Still, these incentives were met with limited success–for example, in its first ten years, only 390 non-Chinese scientists and engineers enrolled in the Thousand Foreign Talents Plan (TFTP) (Zwetsloot).

The commonly cited reasons for foreigners' hesitance to relocate–despite strong financial incentives–tend to be political and cultural. While there is general admiration for Chinese R&D, only 10% of surveyed foreign scientists would consider taking a post in China due to reasons such as the language barrier, China's political situation, and Internet censorship (Zwetsloot). Furthermore, there is a notable difference between Chinese and overseas work culture. Recounting a visit to Beijing, one professor saw the excessive workloads and tight schedules of Chinese academics, and said he would not be able to conduct quality research and offer productive teaching (Wang). Furthermore, foreign academics also feel unfamiliar with China's *guanxi* culture when it comes to procuring grants. *Guanxi* (关系) refers to reciprocity of social capital–in other words, the Chinese competition for grants and financial support for research typically relies on leveraging connections and relationship networks (Wang). Overall, despite zealous efforts by the Chinese government to attract foreign talent, their efforts were met with little success.

2. Unlike other foreigners, Taiwanese engineers and businesspeople saw China as a hub of economic opportunity with relatively low barriers to entry.

In contrast, Taiwanese engineers and researchers were less likely to be affected by the same cultural hesitancy felt by other foreigners–instead, for much of the past two decades, the Taiwanese identity held strong affinity to that of mainland China. For instance, the same *guanxi* culture exists across the strait, and much of the Taiwanese R&D industry shares the same notoriously hustle-and-bustle work culture. Furthermore, the compatibility of identities goes beyond just a shared language–intermarriages between Taiwanese and mainland Chinese became very common as Taiwanese migrants settled down roots in China. With entire Taiwanese families relocating to China, these enclaves began to further integrate into Chinese society. While there was a period in which Taiwanese-style schools were popular for accommodating different educational preferences, these specialized schools faded from popularity as parents began to send their children into local schools. Previous research showed that, when evaluated by the talent attraction index system, cities like Shanghai and Shenzhen were particularly attractive due to their cultural atmosphere and living environment (Hu *et al*). By some estimates, there were over 300,000 Taiwanese people in the Shanghai area by the early 2000s, and continued to steadily increase over the next decade (Leng). Although this Taiwanese community disliked the CCP, they still felt affinity to the historical and cultural elements of the Chinese identity.

During this pre-2019 period, the affinity of Taiwanese and Chinese identities opened the doors to a steady stream of cross-strait economic opportunity. Not only were Taiwanese talent able to take advantage of the wealth of financial incentives offered by Beijing; they were also able to enjoy the open culture and vibrant innovation environments specially architected by the Chinese government. For example, Shenzhen's talent policy was structured not just around boosting basic scientific research, but to cater specifically to "scientific-entrepreneurs" (Wang). In response, many high-tech Taiwanese firms began to establish their own R&D teams in China as a way to leverage both Chinese manpower and market. These firms structured their long-term strategies around the Chinese market as a secondary–or complementary–branch for their businesses to grow. The firms, while physically located in China, mostly only instated high-skill Taiwanese talent as managers and senior-level engineers. Some studies have shown that the mainland branches of these large Taiwanese enterprises were often seen as a "training ground" for the second generation of young business leaders. While the first-generation entrepreneurs stayed in Taiwan, the younger generation may be educated in prestigious Chinese universities and trained from an early age to build *guanxi* networks (Leng). Given the structural accommodations and steady pace of growth, Mainland China was seen as a safe and natural extension of long-term economic opportunity.

At this point in time, Taiwanese firms did not fear a cross-strait brain drain nor risks of espionage. While Taiwan was always wary and protective of its technological advantages, this wariness typically only manifested in high barriers of entry for any high-skilled migrants trying to enter Taiwan. In the 1990s, Taiwan would only allow mainland Chinese researchers to conduct short-term research in Taiwanese institutions. The early 2000s saw slight expansion of allowances, with the Ministry of Economic Affairs supporting individual Taiwanese firms in applying for bringing over advanced workers from China. Still, the proportion of Chinese talent was capped to be at most 10% in any R&D staff (Leng). While restrictions relaxed over the next decade, bringing Chinese talent to Taiwan was still a process riddled with complex review processes and bureaucratic red tape. There was not nearly as much concern for talent flowing in the opposite direction–because most of the Taiwanese talent moving to China were doing so under Taiwanese investment efforts and in managerial roles, the exchange of know-how was seen as mutualistic and advantageous.

Analysis of the Post-2019 Period: Labor shortages and geopolitical events heightened wariness of cross-strait migration.

1. Causes for the post-2019 attitude shift.

In the post-2019 period, however, the attitude towards cross-strait talent migrations have shifted drastically. The shift in attitude comes in part as a result of the drastic shortage of talent in the electronics and semiconductor industries. As the semiconductor industry continues to boom globally, the most significant parts of the manufacturing and design process remain heavily concentrated in Taiwan, Korea, and other Asian countries. Companies such as TSMC have expanded accordingly, accelerating their own hiring rates to meet market demands. According to the Taiwan Institute of Economic Research, the domestic industry employed over 290,000 workers by the end of 2021, compared to the 225,000 workers in 2019 (Cheng & Li). TSMC and MediaTek, the two largest chipmakers on the island, hired over 10,000 employees in 2021, and United Microelectronics aimed to hire 1,500 employees in Taiwan in 2022. A survey by the 104 Job Bank estimated that there were 34,000 vacancies for chip industry positions by 2022, up 77% from 2019 (Cheng & Li).

In contrast to the spike in demand, the supply side struggled to keep up. Over the past decade, Taiwan has been experiencing a decelerating birth rate (Cheng & Li). Some people, like a professor at National Yang Ming Chiao Tung University, believe this to be just a relative mismatch of supply and demand. Others, like a Dean at National Taiwan University, sees this shortage as a major cause for policy urgency for the Taiwanese Executive Yuan. He was watching the telecast in May 2022 when the Executive Yuan passed a new law for human talent cultivation–he said he didn't think such a large legislative effort could have passed six to seven years ago. Whereas the pre-2019 environment would've seen other Taiwanese industries lobbying against such a narrow-scoped investment into just the semiconductor industry, the post-2019 environment saw widespread public support. This shift in attitude paved the way for swift and decisive introduction of new policies from the Executive Yuan.

2. Taiwan passed legislation in response to heightened wariness of Chinese espionage.

In the highly partisan rough-and-tumble of the Taiwanese government, this rare united front stems from heightened wariness of the national security risks posed by China. In 2019, the Executive Yuan passes a series of acts and principles for regulating the country's cyber infrastructure. The "Principles on Restricting the Use of Products That Endanger National Cyber Security" prohibited government agencies from using information and communications technologies (ICTs) that may be security risks–while not explicit in calling out Chinese technologies, steps were then taken nationwide to replace Chinese ICTs and exclude Huawei products (Lee). In 2020, Taiwanese researchers reported that a Chinese hacking campaign may have compromised the internal data of at least seven Taiwanese chip firms (Lee). Since then, TSMC has enhanced its security measures and joined the Taiwanese Industrial Technology Research Institute to adopt best practices and strengthen vulnerabilities. Taiwan also collaborated with the U.S. to co-host a joint Cyber Offensive and Defense Exercise later that year (Lee). Even if individual companies wanted to hire out of China or individuals wanted to work in China, heightened security concerns mitigated easy dissemination of knowledge across the strait.

In accordance with these geopolitical risks, job offers from Chinese companies have since become highly regulated. In 2022, the Act Governing Relations between the People of the Taiwan Area and the Mainland Area (臺灣地區與大陸地區人民關係條例, or the Cross-Strait Act) have received numerous amendments to restrict movement of talent between Taiwan and China. To protect "trade secret involving national core technology," Articles 9 and 91 were amended to 1) establish a review mechanism for personnel engaged in core technologies as they request to travel to China, and 2) establish punishments and penalties for proxy agents that aid mainland Chinese enterprises in illegally conducting business in Taiwan or investing in Taiwan (Mainland Affairs Council). (See Appendix for further details.) The latter of these two amendments specifically target the trend of Chinese enterprises in Taiwan through proxies. There are multiple ways to circumvent detection, either by investing under a different individual's name, concealing their identities, or providing misleading funding sources. The new amendments require any profit-seeking enterprises invested by Chinese enterprises must be pre-approved by authorities; if any were found to be in violation of these measures, they could face fines between NT$120,000 and NT$25 million.

3. Lucrative offers from Chinese firms persisted, but have moved under the table and diminished in recruiting value.

Under this atmosphere of high geopolitical risk, the previously prolific fivefold, sixfold offers from Chinese companies have greatly diminished from publicly available job posting sites. To collect data on what offers are currently publicly available for Taiwanese workers, I created a web scraper for the 104 Job Bank, the main job board in Taiwan. (See Appendix for details.) Out of over 3,000 listings under the Research and Development category, only 58 listings were located in China. Compared to the 60.7% of Taiwanese listings including a publicly visible salary range, only 29.3% of the Chinese listings made the salary range visible. Furthermore, all mainland Chinese listings included the following disclaimer next to its location: "This is a Taiwanese mainland company approved by the country's Investment Review Commission." Interestingly, listings located in Hong Kong do not share this same disclaimer. Given the lack of over-the-table paths toward securing a job in China, the current-day connotation of accepting roles in a Chinese enterprise carries the implication of under-the-table dealings.

Indeed, as numerous cases of engineers breaching espionage laws made headlines, China's offers of high wages and financial compensation became less desirable. Recently, the case of Bitmain, a Chinese cryptocurrency startup, made news for circumventing the National Security Act measures by setting up a Taiwanese engineer as the company chairman. This engineer further recruited former colleagues from a Taiwanese company to form a headhunting team. They posted public listings and offered salaries twofold the industry standard. According to Nanya Technology President Lee Pei-ing, this was in line with the usual strategy of Chinese headhunting efforts–they would first sign on one Taiwanese manager and leverage his or her influence to poach more colleagues (Wu). As such, there is notable social stigma surrounding the topic. When I interviewed professors at prestigious Taiwanese universities, they each rejected the idea of ever accepting any offers from Chinese enterprises. Some gave simple answers about the impracticality of it; most seemed shocked that I would ask such a question, before explaining that such dealings were likely illegal for people in their position.

Furthermore, while most Taiwanese students reported salary to be one of the most important criteria in selecting their post-graduate jobs, they mostly referenced various salaries offered by Taiwanese firms as being sufficiently lucrative. Compared to the NT$25,250 legal minimum monthly wage of Taiwan, the Ministry of Labor found that the average monthly starting salary for a worker in the semiconductor industry to be around NT$52,288 (Cheng & Li). For students graduating from prestigious universities funneling into top firms like MediaTek, the monthly starting salary reaches even higher, to around NT$83,000. Notably, this is just the base salary–the semiconductor industry is known to offer high bonuses. With bonuses included, Intelligent Manpower estimates the annual salary for STEM graduates of top schools to be around NT$1.6 million to NT$2 million (Cheng & Li). My own interviews corroborated these numbers. Graduate students from NTUST and NYCU reported expecting starting salaries to be around NT$1 million, with companies like TSMC and MediaTek offering closer to NT$2 million for IC design roles. Ph.D. students estimate around NT$2 million for their post-graduate industry jobs. Most interviewees considered these annual salaries at domestic firms to be both highly desirable and reasonably attainable, and did not express particularly strong need nor desire to achieve more.

Still, these same Taiwanese students did not completely reject the idea of even more lucrative offers from Chinese firms on the basis of upholding Taiwanese allegiance. Some expressed the same stigmatized reactions as their high-ranking professors–when interviewing a group of NTUST students, one urged me to keep the interview data confidential and anonymous, for fear of retaliation by their more politically opinionated professor. Still, others were open to answering the question of what multiplier threshold would convince them to consider a role at a Chinese enterprise. An NYCU student stated that she would consider any offer over 1.5x the current Taiwan industry average; another NYCU student cited 2x-3x. An NTUST G2 student stated he would only consider offers over 10x, as he felt the usually rumored 5x was not enough to make up for the loss in exposure to advanced machinery. A TSMC engineer confirmed that Chinese companies have offered 5x-6x, but did not feel compelled due to similar reasons for a disruption in long-term career plans.

4. While Chinese roles were wrapped in social stigma, increased polarization of the Taiwanese social identity neither drove workers out of China nor into the Taiwanese semiconductor industry.

Notably, when describing their hesitancy towards working in China, Taiwanese students and workers made no reference toward the cross-strait social identity. As previously discussed, in the 2000s, there was particular mutualism in cross-strait enterprises and R&D due to Taiwanese affinity to Chinese culture, history, and work norms. Where other foreigners cited unfamiliarity with the language, *guanxi* culture, and political environment as reasons not to take lucrative offers from the Thousand Foreign Talents Program, Taiwanese workers and researchers found it relatively easy to integrate into more open-cultured innovation hubs like Shenzhen or Shanghai. More recently, however, surveys by the Election Study Center at National Chengchi University have found increased polarization between the Taiwanese and Chinese social identities. Whereas fewer than 20% of ROC citizens identified as purely “Taiwanese” compared to the 25% that identified as purely “Chinese” in 1992, the same survey in 2016 found that around 60% of ROC citizens identified as purely “Taiwanese” compared to the less than 10% that identified as purely “Chinese” (see Appendix). Although I do not have data on how these students identified, none of the interviewed students made comments that suggested devaluing the Chinese identity (relational content).

Furthermore, there was little to no reference to social identity forces driving Taiwanese students to support their domestic semiconductor industry. Most interviewees acknowledged the complexity of the topic and ascribed statements to be personal preferences. From the interviewed sample, there were no statements on constitutive or relational traits to be applied across all Taiwanese people (cognitive model), or appeal to any maximization of Taiwanese goals (social purpose). There was also no particular praise attributed to those who preferred to stay and work in domestic firms (role theory). I did, however, find that some professors and deans were more likely to make overarching statements about how the blooming semiconductor technology in

Taiwan and high domestic pay should incentivize most engineers to stay on the island. On the other hand, still more university faculty members boasted about their program's offerings of international experiences.

Instead of particular interest in contributing to the Taiwanese semiconductor industry as role theory would suggest, most students actually expressed interest in either going abroad or working for a foreign firm (外商), citing a better work-life balance. Work-life balance was the top criteria for choosing post-graduation jobs, following after financial compensation. The Taiwanese industry is infamous for its strenuous workload and "liver-selling" (賣肝) culture. One student described reasonable working conditions to be less than twelve hours per day, and another cited that his older brother and father experienced regular overtime to make deadlines for a project. A TSMC engineer explained that most workers had no actual holidays or weekends, and recent graduates often have to work "small night" or "big night" shifts, both of which had hours stretching in the AM. In addition to the clocked hours, one student also pointed out TSMC's protection measures that strictly regulated an employee's movement and knowledge within the company. In strong contrast to the open-office concepts of Facebook or Google, TSMC does not allow employees to wander beyond a limited section of its office. While students may compete for more upstream IC design roles, which are known to be less strenuous, foreign firms are still the best option for those who would like to work reliable eight-hour days with good pay. Several students cited that, despite the robust promotion system and reliable pay raises (a TSMC engineer reports that, after 7 years in the company, his salary has over-doubled from NT$1.1 million to NT$2.4 million), they did not find it sustainable to stay in industry for over five to ten years.

Notably, despite many interviewees expressing interest in going abroad, it is unclear whether the desire to work overseas involves short-term experiences or long-term relocation. An NYCU postdoc researcher recalled his time in Germany, where the company required very low working hours (around half of Taiwan's) since contracts preferred projects to be settled and well-tested prior to completion–still, he returned to Taiwan to further pursue his research career. Companies like MediaTek also offer positions at their international branches in Singapore and other locations. An NTU dean noted that the younger generation demonstrates greater desire to work globally, as they are more willing to relocate to roles in Japan, Singapore, Australia, and the U.S. Under this globalizing philosophy, the school conducted the majority of graduate courses in English and encouraged students to gain experiences from positions abroad. The same dean was not particularly concerned about a potential brain drain–he was proud in describing NTU's efforts in emphasizing an international mindset in their curriculum.

5. Geopolitical factors made Chinese firms lackluster in the skills development and stable career trajectory that high-skill talent sought after.

When discussing long-term goals, many students also expressed interest in developing their skills, which both drew them away from the lagging Chinese industry and toward research positions in both Taiwan and abroad. Due to the geopolitical environment, many of the cutting-edge technologies critical for the semiconductor R&D and manufacturing processes are not available in China. For example, extreme ultraviolet (EUV) lithography, a process that can shrink billions of transistors on a single 2-nanometer chip (Murphy), can only be done using the $150 million machine produced by the Dutch company ASML (Knight). Only a few handful of companies in the world can afford the ASML EUV machines, including TSMC, Samsung, and

Intel. In 2019, in a geopolitical effort to bottleneck China's semiconductor industry, the U.S. persuaded ASML to block shipments of the most state-of-the-art machines to China (Swanson *et al*). As ASML continued to look for profit off lucrative sales to China, both the Trump administration and the Biden administration continued working to further curb sales of related machinery and tighten export controls. Other machine manufacturers, such as Applied Materials and Lam Research, have also come under similar export controls regarding sales to China.

In addition to blocks on physical machinery, Washington has also instituted multilateral export controls on distributing electronic design automation (EDA) software licenses to any non-U.S. ally (Yang). The top three EDA companies–Cadence, Synopsys, Mentor Graphics–are American or American-originated, and form an oligopoly capturing 70% of the global EDA market. Manufacturing companies often require software tools from these top companies to design the most advanced circuit structure–called the gate-all-around field-effect transistor, or GAAFET (Yang).

As a result of these geopolitical export controls, the 2023 Chinese semiconductor manufacturing capacity is still bottlenecked by EDAs capable of designing 5mm chips and a very limited supply of EUV machines. Therefore, for students, researchers, and engineers looking for a stable career trajectory, taking roles in Chinese firms may lead to risks of critical lags in skill development down the line. In contrast, Taiwanese companies offer more opportunities of lateral or vertical movement. For example, some students cite that opportunities for IC design roles at MediaTek or Qualcomm may come more easily after experience in design or foundry work at TSMC. As such, despite the lure of short-term financial benefits offered by Chinese roles, staying in the domestic industry paves the way for a stable and predictable career trajectory.

6. The Taiwanese academia-industrial complex adds resiliency against geopolitical factors and career stability for semiconductor talent.

In further contrast to the more unpredictable trajectory of the Chinese semiconductor industry, the Taiwanese government has also invested substantially in creating a robust academia-industrial complex. An NTU dean described how the government's new funding-matching program: should a school find a collaborative company to sign on to committing a certain amount of funding over a long-term period, the National Science and Technology Council (NSTC) will match said funding. For NTU, much of the recent funding for semiconductor research stems from a government-matched collaboration with TSMC, PowerChip, MediaTek, and Etron ("Joint Research Centers with Industry Leaders"). From the government perspective, the Director General of the Department of Academia-Industry Collaboration and Science Park Affairs at NSTC described the strategy behind university-industry collaborations: when companies come across challenges they struggle to solve, they leave the challenges for universities to solve. She gave the example of TSMC–the large manufacturer has four separate project plans across NTU, NTHU, NYCU, and NCKU. NTU is tasked with making smaller chips and developing the 1-nanometer manufacturing procedure; NTHU is tasked with developing EUV equipment; NYCU is tasked with researching more energy-efficient materials for future CPUs. Other companies like China Steel or Chang Chun Petrochemicals collaborate with universities for developing green technologies.

This academia-industrial complex provides Taiwanese talent with a direct pipeline into a stable career. For one student, his lab's specialty was in device simulation, and he knew it an area of skill that would best transfer well into TSMC roles. Additionally, NYCU has a natural proximity to TSMC that helps facilitate greater collaboration. Many of NYCU's labs are signed onto industry-requested collaboration projects with the university. When asked about the career center, a professor also recalled an anecdote where much of the recruiting happens when alumni drop by the lab and informally hire to-be-graduates then and there. Since the skillset is so specialized, the alumni network is focused, vast, and strong, and helps guarantee top students direct lines to jobs in the industry. It should be noted that many labs in Taiwan, not just those in NYCU, share these characteristics. All semiconductor programs have collaboration programs with universities, and sponsoring companies often provide scholarships and paid research programs. An estimated 80% of semiconductor-related students in universities are engaged in some way, either through scholarship or through funded research projects. Furthermore, when working on these funded projects, students are quickly exposed to routine meetings with professionals from the industry to get specifications and requirements for the product. As such, Taiwanese talent have access to a very robust pipeline of career opportunities in the domestic industry, thus further disincentivizing talent to accept offers in Chinese enterprises.

Conclusion

Our hypothesis testing found that short-term financial compensation in the form of high salaries, bonuses, and other monetary perks were effective in attracting Taiwanese talent towards Chinese firms in the absence of other drivers, but was ultimately secondary to the geopolitical risk factor. Under increased geopolitical tensions, Taiwanese government crackdowns on illegal headhunting activities by Chinese enterprises resulted in few publicly available listings, with most of the same lucrative offers having gone under the table. As a result, most either view the famous fivefold, sixfold Chinese offers with either deep suspicion of illegality or with only hypothetical consideration. Individuals express different attitudes in weighing the importance of higher salary, but most were generally satisfied with the financial compensation offered by Taiwanese companies. As such, we see that short-term financial compensation is overshadowed by geopolitical risk considerations.

Furthermore, despite increased polarization of the Taiwanese identity, identity neither drives talent out of China nor toward Taiwan. While there have been shifts in identity, with many Taiwanese people increasingly emphasizing elements of democratization and de-Sinification, along with general social pressure surrounding the stigma of working for Chinese firms, the parts of identity most relevant to determining migration remain unchanged. Taiwanese identity continues to share norms with the Chinese identity, as shown by high intermarriage rates and *guanxi* culture. Furthermore, many Taiwanese talents highly prioritize work-life balance, and would prefer to work for less strenuous foreign firms if given the opportunity.

Finally, in the face of geopolitical risk and instability in the semiconductor industry, Taiwan's strong academia-industrial complex and access to the world's cutting-edge tools provide stability for a worker's career. Few Taiwanese high-tech firms continue to view China as an opportunistic market to enter. Despite more lucrative short-term compensation, many Taiwanese individuals cite economic uncertainty and potential back-step in skill development as reasons to reject Chinese offers. While annual salaries in China may be higher, workers fear the lack of access to EDA software and EUV lithography as a result of U.S. export controls would irreparably hinder their long-term career trajectory. In contrast, strongly government backed and *guanxi*-fueled university-industry collaborations often secure top Taiwanese talent in engineering roles even before they graduate.

Moving forward, the Executive Yuan should keep in mind the effectiveness of its policies relating to the Taiwanese academia-industrial complex, along with the competitive advantage it currently holds in the cutting-edge of semiconductor innovation. To tackle the severe labor shortage, it may be worth considering increasing efforts to scout talent from Southeast Asia. While there already exist programs that focus on bringing high-skill talent from Indonesia and other ASEAN countries, Taiwan may benefit from further liberalizing labor mobility in this region. Given that few other places in the world may offer the same career stability, financial compensation, and long-term skills development, Taiwan may have more leverage than expected in keeping top engineering talent in Hsinchu. Such policy proposals may require further investigation and corroboration, but suggest that Taiwan still holds some strong bargaining chips in the complex entanglement of the global semiconductor industry.

Limitations and Next Steps

This paper faced major limitations in the lack of aggregate data and peer-reviewed statistics on migration trends and industry funding. In addition to the lack of workforce data, I was also unable to find publicly available NTSC digital yearbooks for the figures on how much aggregate funding the Taiwanese government has invested in its academia-industry complex. I was also unable to source Chinese job listings to compare against the data I scraped from the 104 Job Bank. Furthermore, when interviewing critical stakeholders, individuals expressed greater reluctance to talk than expected. Given the highly political nature of the topic, many of these data points may be difficult to accurately procure in the coming years–nevertheless, the most important next steps would be to find alternative ways to estimate such metrics.

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Appendix

1. Web scraper for the 104 Job Bank.

The code is publicly available at: https://github.com/JessSanChen/104scraper

1. Used Selenium to dynamically load page (access page numbers)
2. Used Beautifulsoup to grab job listings on any page, given search URL
3. Search parameters:
    a. Must manually create Taiwan URL query since web UI disallows the selection for > 10 counties
        i. Only search for 研發相關類 (R&D), no 資訊軟體系統類 (IT) for now
    b. Salaries only accessible on the listing's webpage; must create separate soup request to retrieve each
4. Used Pandas DataFrames to aggregate data
    a. Columns: "'job_name', 'job_url', 'location', 'area', 'company', 'education', 'experience', 'role_types', 'salary', 'shift_time', 'role_term', 'skills'
5. Output as CSV file

	job_name	job_url	location	area	company	education	experience	shift_time	role_term	skills	role_types	salary
2743	Opto-Mechanica	http://www.104.c	Taiwan	台北市南港區	新析生物科技股	大學	5年	日班	全職		機構工程師,光電	年薪1,000,000~1,700,000元（固定或變動薪資因
1072	應用工程師	http://www.104.c	Taiwan	台北市信義區	安矽思科技股份	大學	1年	日班	全職	ANSYS,EDA	FAE工程師,電子	年薪1,000,000元以上（固定或變動薪資因個人資
1537	材料開發專員	http://www.104.c	Taiwan	台北市松山區	格雷蒙集團_台灣	大學	2年	日班，09:00~18	全職	英文,Excel,Interr	國內業務,國外業	年薪1,000,000元以上（固定或變動薪資因個人資
2422	[2024正職及預聘	http://www.104.c	Taiwan	台北市松山區	德州儀器工業股	大學	不拘	日班，09:00~18	全職	英文	電子工程師,業務	年薪1,000,000元以上（固定或變動薪資因個人資
2461	[2024正職及預聘	http://www.104.c	Taiwan	台北市松山區	德州儀器工業股	大學	不拘	日班，09:00~18	全職	英文,PCB Layou	電源工程師,硬體	年薪1,000,000元以上（固定或變動薪資因個人資
2956	電源工程師 Pow	http://www.104.c	Taiwan	台北市南港區	德倫思管理顧問	專科	2年	日班	全職		電源工程師,硬體	年薪1,200,000~1,500,000元（固定或變動薪資因
124	Technical Head/	http://www.104.c	Taiwan	台北市內湖區	Affiliates.One 聯	大學	5年	日班	全職	英文,Github,Rub	硬體工程研發主	年薪1,500,000~2,000,000元（固定或變動薪資因
468	外商電子代工-機	http://www.104.c	Taiwan	台北市信義區	Spring Professio	大學	8年	日班，9AM to 6	全職	英文	生產技術 / 製程	年薪1,500,000~2,200,000元（固定或變動薪資因
469	外商電子代工-機	http://www.104.c	Taiwan	台北市信義區	Spring Professio	大學	8年	日班，9AM to 6	全職	英文	生產技術 / 製程	年薪1,500,000~2,200,000元（固定或變動薪資因
470	外商電子代工-機	http://www.104.c	Taiwan	台北市信義區	Spring Professio	大學	8年	日班，9AM to 6	全職	英文	生產技術 / 製程	年薪1,500,000~2,200,000元（固定或變動薪資因
1647	LC-Power徵才專	http://www.104.c	Taiwan	台北市信義區	新加坡商立可人	大學	10年	日班	全職	英文	硬體研發工程師,	年薪1,500,000~3,500,000元（固定或變動薪資因
729	Chief Scientific	http://www.104.c	Taiwan	台北市南港區	新析生物科技股	博士	不拘	日班	全職	英文	生物學研究員,生	年薪1,800,000元以上（固定或變動薪資因個人資
89	JL - [全球500大	http://www.104.c	Taiwan	台北市內湖區	新加坡商立可人	大學	不拘	日班	全職	英文	FAE工程師,數位	年薪2,000,000~3,000,000元（固定或變動薪資因
2479	實驗室專員/助理	http://www.104.c	Taiwan	台北市內湖區	漢康生技股份有	專科	不拘	日班	全職	Excel,Outlook,P	醫藥研發人員,生	年薪420,000~530,000元（固定或變動薪資因個
2975	『台北辦公室』	http://www.104.c	Taiwan	台北市大安區	德瑞精密機械有	不拘	不拘	日班，08:30~17	全職	英文,機械產品操	助理工程師,工程	年薪460,000元以上（固定或變動薪資因個人資歷
652	Hardware Mana	http://www.104.c	Taiwan	台北市內湖區	艾知科技股份有	大學	不拘	日班	全職	英文	硬體工程研發主	年薪500,000~1,800,000元（固定或變動薪資因
649	UAV Engineer無	http://www.104.c	Taiwan	台北市內湖區	艾知科技股份有	大學	不拘	日班	全職	英文,產品結構評	機構工程師,熱傳	年薪500,000~780,000元（固定或變動薪資因個
1676	農業計畫管理與	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	碩士	1年	日班，8:30~09:1	全職	英文,Excel,Powe	其他研究人員,病	年薪500,000~800,000元（固定或變動薪資因個
2788	Tissue Scientist	http://www.104.c	Taiwan	台北市南港區	新析生物科技股	大學	3年	日班	全職		生物科技研發人	年薪500,000~800,000元（固定或變動薪資因個
2760	Scientist, Cell Bi	http://www.104.c	Taiwan	台北市南港區	新析生物科技股	碩士	2年	日班	全職	英文	生物科技研發人	年薪520,000~1,050,000元（固定或變動薪資因
171	3動物植物防檢疫	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	農藝 / 畜產研究	年薪550,000~800,000元（固定或變動薪資因個
179	4林業綱要計畫專	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	農藝 / 畜產研究	年薪550,000~800,000元（固定或變動薪資因個
93	專案管理師	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	其他專案管理師,	年薪550,000~850,000元（固定或變動薪資因個
162	政策溝通研究員	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	其他專案管理師,	年薪550,000~850,000元（固定或變動薪資因個
448	策略規劃與產業	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	其他專案管理師,	年薪550,000~850,000元（固定或變動薪資因個
492	產業培力顧問	http://www.104.c	Taiwan	台北市大同區	社團法人台灣農	大學	不拘	日班，08:00~09	全職	英文,Excel,Powe	其他專案管理師,	年薪550,000~850,000元（固定或變動薪資因個
2488	機電設計工程師	http://www.104.c	Taiwan	台北市大安區	國泰建設股份有	專科	不拘	日班，08:30~17	全職	英文,AutoCAD	機電技師 / 工程	年薪560,000~1,200,000元（固定或變動薪資因
680	日本硬體工程師-	http://www.104.c	Taiwan	台北市中山區	安卓美達科技有	大學	5年	日班	全職	MCU,Circuit Des	電子工程師,零件	年薪600,000~1,000,000元（固定或變動薪資因
267	CAE 結構模擬應	http://www.104.c	Taiwan	台北市中山區	艾索科技股份有	碩士	不拘	日班	全職	英文,Adobe Acrc	機械工程師	年薪650,000~1,000,000元（固定或變動薪資因
314	CAE 流體模擬應	http://www.104.c	Taiwan	台北市中山區	艾索科技股份有	碩士	不拘	日班	全職	英文,Adobe Acrc	機械工程師	年薪650,000~1,000,000元（固定或變動薪資因
1736	CAE 5G通訊、	http://www.104.c	Taiwan	台北市中山區	安捷新科技股份	大學	不拘	日班	全職	英文,Adobe Acrc	電機技師 / 工程	年薪650,000~1,000,000元（固定或變動薪資因
1886	CAE 低頻馬達應	http://www.104.c	Taiwan	台北市中山區	安捷新科技股份	大學	不拘	日班	全職	英文,Adobe Acrc	電機技師 / 工程	年薪650,000~1,000,000元（固定或變動薪資因
1903	CAE 應用工程師	http://www.104.c	Taiwan	台北市中山區	安捷新科技股份	大學	不拘	日班	全職	英文,Outlook,Po	機械工程師	年薪650,000~1,000,000元（固定或變動薪資因
2425	熱流、聲學CAE	http://www.104.c	Taiwan	台北市中山區	士盟科技股份有	碩士	不拘	日班，08:30~17	全職	英文,ABAQUS,A	聲學 / 噪音工程	年薪650,000~1,000,000元（固定或變動薪資因
2350	結構CAE工程師	http://www.104.c	Taiwan	台北市中山區	士盟科技股份有	碩士	不拘	日班，08:30~17	全職	英文,ABAQUS,P	機構工程師,機械	年薪700,000~1,000,000元（固定或變動薪資因
2291	CAE軟體二次開	http://www.104.c	Taiwan	台北市中山區	士盟科技股份有	大學	不拘	日班	全職	英文,Linux,ABA	軟體工程師,機構	年薪700,000~1,100,000元（固定或變動薪資因
446	Hardware Desig	http://www.104.c	Taiwan	台北市南港區	恆暘股份有限公	不拘	3年	日班，09:00~18	全職	MCU,RF,PCB La	硬體研發工程師,	年薪700,000~1,600,000元（固定或變動薪資因
1114	醫學編輯(Medic	http://www.104.c	Taiwan	台北市大安區	法米迪亞醫學有	大學	不拘	日班	全職	英文,文書處理軟	病理藥理研究人	年薪750,000~1,000,000元（固定或變動薪資因
218	售服機電主管	http://www.104.c	Taiwan	台北市大安區	寶舖建設股份有	專科	5年	日班	全職	Excel,PowerPoir	電機技師 / 工程	年薪800,000~1,000,000元（固定或變動薪資因
144	資深醫學編輯(S	http://www.104.c	Taiwan	台北市大安區	法米迪亞醫學有	大學	不拘	日班	全職	英文,文書處理軟	病理藥理研究人	年薪800,000~1,200,000元（固定或變動薪資因
1545	資深醫學編輯(M	http://www.104.c	Taiwan	台北市大安區	荷陽創意行銷整	大學	2年	日班	全職	英文	文編 / 校對 / 文	年薪800,000~1,200,000元（固定或變動薪資因
2742	Mechanical Engi	http://www.104.c	Taiwan	台北市南港區	新析生物科技股	大學	2年	日班	全職		機構工程師,光電	年薪800,000~1,400,000元（固定或變動薪資因
1124	設計部副理(電氣	http://www.104.c	Taiwan	台北市中正區	大桂環境科技股	專科	3年	日班	全職	Adobe Acrobat,E	電機技師 / 工程	年薪800,000元以上（固定或變動薪資因個人資歷
362	硬體研發資深工	http://www.104.c	Taiwan	台北市內湖區	南京資訊股份有	大學	不拘	日班，9:00~18:0	全職	英文	電子工程師,硬體	年薪850,000~1,200,000元（固定或變動薪資因
1645	資深 CAE 工程	http://www.104.c	Taiwan	台北市中山區	安捷新科技股份	碩士	5年	日班	全職	英文,Outlook,Po	機械工程師,熱傳	年薪900,000~1,260,000元（固定或變動薪資因
58	【建設部】資深	http://www.104.c	Taiwan	台北市中山區	香城飯店企業集	專科	3年	日班，需輪班	全職	Excel,Word,Autc	土木技師 / 工程	待遇面議（經常性薪資達 4 萬元或以上）
60	室內裝修專案工	http://www.104.c	Taiwan	台北市大安區	普世康生技室內	高中	5年	日班	全職	台語,AutoCAD,A	工務人員 / 助理,	待遇面議（經常性薪資達 4 萬元或以上）
61	【建設部】機電	http://www.104.c	Taiwan	台北市中山區	香城飯店企業集	專科	5年	日班	全職	Excel,Word,Autc	機電技師 / 工程	待遇面議（經常性薪資達 4 萬元或以上）
62	Software Engine	http://www.104.c	Taiwan	台北市信義區	Google_美商科	不拘	不拘	日班	全職		硬體研發工程師,	待遇面議（經常性薪資達 4 萬元或以上）
63	Platform Therma	http://www.104.c	Taiwan	台北市信義區	Google_美商科	不拘	不拘	日班	全職		硬體研發工程師,	待遇面議（經常性薪資達 4 萬元或以上）

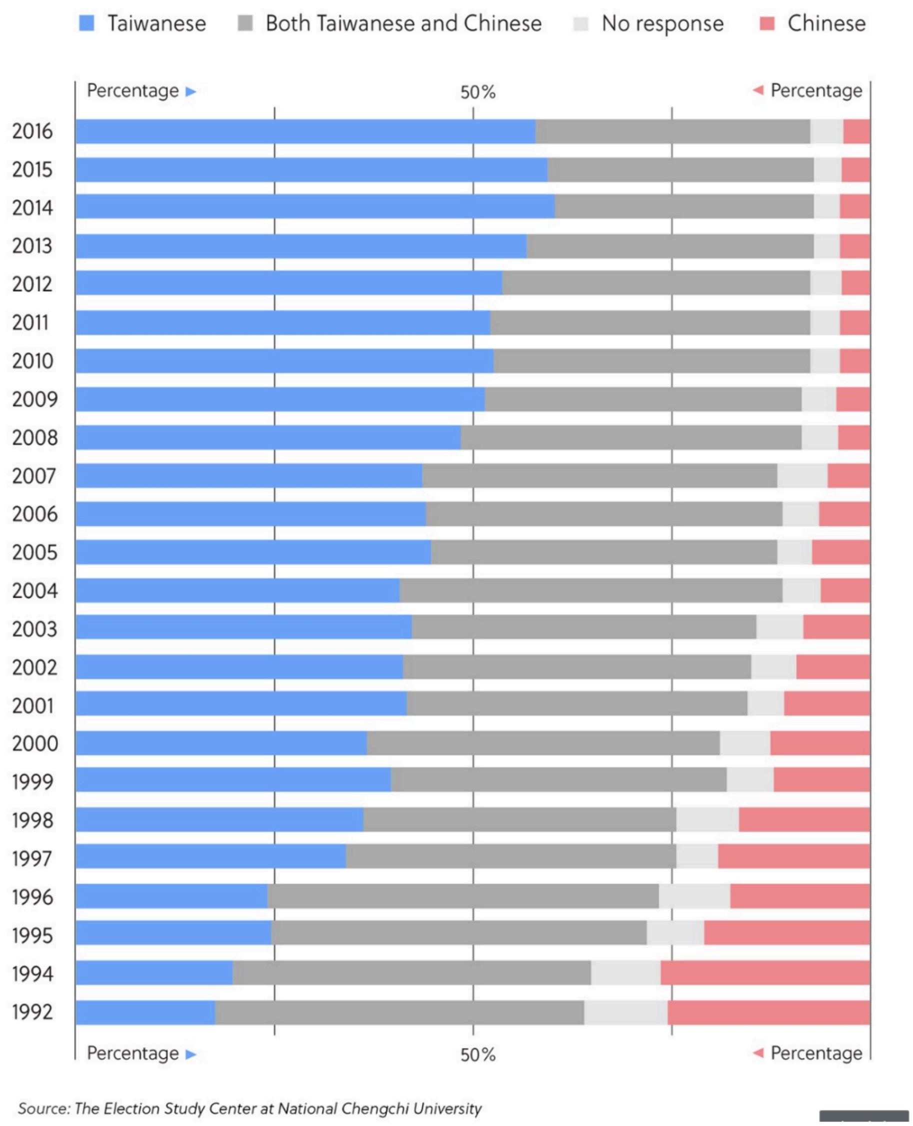
2. Amended articles in the Act Governing Relations between the People of the Taiwan Area and the Mainland Area

Source: Mainland Affairs Council. *Act Governing Relations between the People of the Taiwan Area and the Mainland Area* , Laws & Regulations Database of the Republic of China (Taiwan), 2022.

The People of the Taiwan Area with the following status shall apply for permission to enter into the Mainland Area. The application shall be reviewed and approved by a committee formed by the Ministry of the Interior, together with the National Security Bureau, the Ministry of Justice, the Mainland Affairs Council, and the relevant agencies… Individuals or members of legal persons, organizations or other institutions entrusted, subsidized or invested to a certain level by government agencies (institutions) who engage in matters involving national core technologies. (Article 9)

Any person of the Taiwan Area with the status as referred to in Subparagraphs 3, 4, or 6, Paragraph 4 of Article 9 who violates the provisions of Paragraph 5 of Article 9 may be punished by the (original) agency where such person serves/served or the entrusting, subsidizing, or investing agencies (institutions) with an administrative fine of not less than twenty thousand but not more than one hundred thousand New Taiwan Dollars.
Any person who violates the provisions of Paragraph 8 of Article 9 pertaining to a failure to make the required filing may be punished by the (original) agency where such person serves/served with an administrative fine of not less than ten thousand but not more than fifty thousand New Taiwan Dollars. (Article 91)

<u>3. Increasing polarization between the Taiwanese and Chinese identities.</u>
Source: The Election Study Center at National Chengchi University.



<u>4. Compiled raw notes from interviews.</u>

- What programs or opportunities does your lab offer to students in preparation for their job search?
    - NTU Dean Tzi-Dar Chiueh
        - Heavy emphasis on international and practical mindset
        - First academic year implemented changes to requirements:
            - Take mandatory course: “internship”
                - 3-credit course, 8 weeks in industry, preferably in the summer
            - Take English as medium of instruction courses
                - Undergrad: 80-90% courses are in Mandarin, very low in English
                - Graduate: >50-60% in English
            - Invite industry practitioners to give course modules, practical experience in all 3 areas
        - 3 areas of specialty
            - IC design, automation: electronic design, software program
            - Semiconductor devices: fabrication, advanced manufacturing
            - Nano-engineering: more fundamental engineering and MechE, chemical, civil engineering, guide students
- Which companies do your students tend to choose after graduation?
    - NTU Dean Tzi-Dar Chiueh
        - Website listed some internship opportunities, but advisors work more one-on-one to encourage students to search more
        - Academica Sinica, MediaTek
        - Goal: more international opportunities (MediaTek offering positions in Singapore and elsewhere)
        - Sees number of students who become professors as good indicator for success of their program, which is interesting
    - NYCU Prof. Becker Hsu
        - 90% go into industry after graduation, 10% go into higher education in PhD
            - Some Masters → PhD stay in NYCU, others go abroad (Japan, etc)
            - Out of the PhD students: 95% go into industry
        - TSMC is always the first choice, high demand due to lack of engineers
            - 99% go into TSMC
        - Special relations to TSMC:
            - Location proximity: TSMC and NYCU
            - All semiconductor programs have collaboration programs with university
                - Scholarships, research programs, mentor programs
                - 80% of students in universities are engaged in some way (scholarship, research projects funded)
    - NYCU Tom (Becker’s student)
        - Post-doc, already graduated
            - Master’s from UCLA, PhD from NYCU and Germany
            - Currently a postdoc at NYCU
        - Much prefer research and faculty position over time in the industry, because of high workload and extreme working environment in Taiwan

            - Industry doesn't require PhD and generally very few professorship positions, so not the most popular option.
            - Industry is good for money
            - But he's already saved some money during graduate studies, so less pressing to make money
        - Lifestyle of academic work:
            - Professorship gives a lot of freedom, don't need to be in lab for 8 hours or 12 hours
            - Don't have a firm working schedule, still need to stick to procedures and schedules for research though
            - Sometimes in university need to work on weekends, but sometimes have free time on weekdays
            - Weekdays mostly just routine management stuff
        - "Best option" career and company for friends going into industry:
            - One member wants to go to TSMC, other member doesn't (his wife works at TSMC, but sees her struggling 8am-8pm work days)
            - ITRI: similar salary to the university, don't need to stress too much, just work on ongoing projects, seek some external funding
                - Research institute in Taiwan: between university and industry
            - NYCU has many collaboration projects with industry
                - Routine meetings to get specifications and requirements for products
                - Develop out of these wants, implementation and design techniques
            - When working in Germany:
                - Collaboration is quite different, since collaboration is made for sale
                - Working hours quite low in Germany (about half of Taiwan's!), not rushing for anything, want everything settled and in good shape
        - Jobs abroad
            - Had some opportunities in Canada Vancouver, but maybe farther in future; right now just want faculty position in Taiwan
                - 10 years later, go to North America industry
                - → younger brother is there, so want to join family there
    - NYCU Gwen (Becker's student)
        - Apply to job!
            - She has applied to several study abroad programs: switzerland, spain, sweden; currently in 2nd year Master's
        - Want to go to TSMC; very similar to what they're already doing
            - Device simulation
        - Some other options: 設備商(KLA)
            - Last year insight program, which can explore many countries abroad
            - They supply the equipment, different company

- NYCU Oliver (Becker's student)
    - 3rd year PhD student
    - If graduate from PhD, then 2 ways:
        - stay in Taiwan, go to company like MediaTek, TSMC, RealTek, big company in Taiwan Hsinchu Science Park
            - Biggest thing: not too much challenge, just stay in familiar environment of Hsinchu
            - Money earned is a lot for Taiwan
            - But just an engineer, do the same routine things every day
        - go abroad, America or Europe for post-doc research
            - Many things to solve: where to live, when can go back to Taiwan
            - But it is something he wants to do, it must be more interesting!
            - Research very encourages him to do more things
            - → also must consider whether to stay abroad for a long time or come back to Taiwan as a professor or something else
            - Will have a decision when almost graduated from PhD (at least 1-1.5 year)
            - → not that easy to find, because must apply for "good" university labs, chance not necessarily a lot; not an easy way to go, challenge
- NYCU Dale (Becker's student)
    - Go into industry
    - Maybe TSMC first, these experience is very good for career
    - Can then go to foreign companies in either in Taiwan or abroad
        - Work-life balance! TSMC has to work very hard, high pressure
        - Other companies, esp from other countries, treat people better in terms of hours, pay, benefits
    - Best option career: Most want to go to TSMC or Mediatek first, but unsure with many want to go to other companies after TSMC
        - Many people in lab recently stay in academic field
        - Usually not that many go through this path, but more in this lab, around 60% (2-3 guys)
        - In terms of PhD students, he's the only one that wants to go into industry; this is unusual
        - → Prof. Hsu chose people who are very passionate about research, so get a lot of resources, so professor helps, gets a lot of happiness from research
- NTUST G2-1
    - 研發替大雨
    - Company give 4 months, only train for a couple days in the military
    - Avoid the military, substitute the mandatory service with over a year in the industry
    - Guys need to do 4 months, but because it's specialized, but service the company instead, commit to the company

                - Identity as a military service, but service
            - Must apply to attain this status, at least 20-30 companies
            - Company work with the government to have these openings, based on availability
                - Government
            - After doing the thesis, need to find job
            - Most people want to try to apply for this, more and more people are interested, since military service
            - Very competitive
        - NTUST G2-2
            - Gap year! Get exposed to different fields under EE/CS
            - Also throw in some resumes
            - Not many people want to pursue this gap year
            - “Best option”
                - In semiconductor industry:
                    - Up: IC design
                    - Middle: TSMC, jingyuanzhizhao
                    - Down: packaging
                    - → want to go to “upstream” which is IC design
                    - Within IC design
                        - Analog → preferable
                        - Digital
                - Want to do IC design analog because it’s closest to the content of the lab
                    - Will be different from lab to lab
                - The more upstream, the more prestigious
                - Best companies: NVIDIA, Intel, etc
                    - Mediatek, Novatek, Realtek, Reachtek
                    - → all upstream to IC design
                - Middle: TSMC, UMC, Global
                    - Fabs
                - Bottom: packaging, not much relation to this
        - TSMC
            - Back when he was graduating, nobody wanted to go to TSMC
            - To be a process engineer, needed to take shifts at bad times at “small night”
            - Other type of engineer: needed to take shifts at “small night” and “big night”
            - No actual holidays or weekends
            - Young people wanted more freedom
            - TSMC is a place where seniors encourage young people to go
            - Personally, still chose because of the money, needed to work harder when young to give himself more of a buffer
            - Professors don’t want
- Criteria for choosing a job/career after graduation
    - NYCU Tom (Becker’s student)
        - Factors in order: working hours, salary, location

        - Hsinchu is purely engineering, not as casual and rushing in Taipei (different pace of life)
    - NYCU Gwen (Beckers's student)
        - Work hours: don't want to work over 12 hours a day, have to have a good environment, not too straightforward (can work at home or at the company)
            - Good environment: TSMC has very strict rules, stay in specific areas; whereas Facebook and Google has more free space
    - NYCU Gwen (Becker's student)
        - Money is not high priority, but still need some money to live
            - Salary: 2,000,000 NTD per year is enough
        - Good devotion on research: interested in new things and research, if chance to explore new things and new technology, or professor/researcher position → is his goal
            - Engineer vs. professor/researcher is quite different
            - What he does in the field he chooses is the most important; salary is not top priority
        - professor/researcher is a big goal
            - If start with industry work, then after 5-10 years switch over to professor/researcher
            - But for now, still want to decide if he wants experience in company work at all
    - NYCU Dale (Becker's student)
        - For first company: salary is top priority
    - NTUST G2-1
        - IC design
        - Work-life balance
            - Don't know much about the real life of working, it's very different from the lab work environment
            - Still figuring out for himself what the best lifestyle (has more flexibility), hard to figure out, only know the job description, will have to enter the job to see
    - NTUST G2-2
        - Development for skills for the future! Most important, experiences that carry over
        - How competitive the company is, top in the world
        - First job, the work-balance not as important
        - Work-life balance:
            - Based on older brother and dad, Taiwanese work environment not as great
            - Often overtime to 7pm or 8pm to make the deadlines for project
            - Based on responsibilities, if you get things done
            - Naturally get less as more senior
- Salary
    - NYCU Tom (Becker's student)

        - Postdoc regular salary of 60,000 NTD per month + bonus at the end of the year
        - Extra bonus from university by publishing outstanding works, patents file from companies
        - "Salary is quite enough": colleague from graduate studies at TSMC, also with PhD, monthly salary 65,000 NTD
            - Working load is quite different, salary not that big of a difference
    - NYCU Gwen (Becker's student)
        - Out of graduation: up to 60,000 NTD a month, 1mil a year
        - After 5 years: go up 20%
        - Does not want to stay in industry for a long time, mostly from parental influence, want to be in the vet instead
            - "Everybody wants to leave"
            - In this neighborhood, everybody's parents are in the industry, usually the father stays in their entire life
            - 80% stay, usually stay unless they get fired
    - NYCU Oliver (Becker's student)
        - If stay in company in Hsinchu: high technology, makes a lot of money, engineer also makes a lot of money
        - Post-doc research: money not so much, value is in experience in teaching, explore new eras and things
        - → after time: if start with 2mil after entering company
            - 5 years: 3-4 million NTD
            - 10 years: 4 million NTD
            - Though performance/devotion also matters for whether you can earn that money or not
        - Promotion system at TSMC:
            - If graduated from master's, get level 31
                - After 2-3 years, get promoted to 32, but then 33 is difficult for master's degree to obtain
            - If graduate from PhD, get level 33 when you enter
                - After 2-3 years, get promoted to 34, but then 35 is management level, which depends on if you want to stay as manager
                - If you prefer to stay on routine work as an engineer, then don't need to promote to 35
            - 50-50 of staying all the way to management
                - Managers have more responsibilities to things and staff
            - For the 50% that don't go into management:
                - 70% stay as routine work as engineer
                - 30% want to find new things to do
                    - Switch to another company, or switch to another department (ex: operations for taking care of equipment, to R&D, switch back and forth)
    - NYCU Dale (Becker's student)
        - 2,000,000 NTD a year for recent graduates with PhD at TSMC

        - For 外商: less working hours (or equal to 8 hours a day)
            - Estimate after 5-10 years to transition
            - Estimate salary: 3-4 million NTD
        - Most people choose to leave TSMC
            - They want young people who have the energy to do the work, because they want to be the best in the world
            - More than 50% of people want to leave after 5-10 years because of life, family
            - Few people stay more than 10, 15, 20
    - NTUST G2-1 and G2-2
        - 1,000,000 NTD per year
            - meditaTek: 2,000,000 NTD
            - TSMC: 1.6 to 2 million
            - Most IC design
        - Salary hierarchy: Designer gets the most money, foundries (work with the companies), then testing
            - 10,000-20,000
        - When chinese hire, it could’ve been 5x salary a couple years ago, but as they have the knowledge transfer, they’ll start hiring higher and higher up the chain, since they need more top-level knowledge
        - There is a large change in knowledge within just a couple years, so they don’t really hope to still be the best engineer in 10 years, they transfer over to a more managerial role
            - Use experiences to do management, can’t be substituted easily
            - Most go through this process of engineer to management
        - Not a huge change in salary, but enough to retain
            - Depends on the specific company, some are more stable, others are more top/bottom
            - IC design and CPU: high tech, big changes as you go up
            - Versus power supply: just stable, not very dynamic, salary reflects this stability
    - TSMC
        - When first entered, monthly salary of NTD 45,000
            - Monthly salary is not that much, but the annual salary is a lot, because lots of bonus, the bonus tends to be as much (if not more) than the base salary
            - Base salary is 14 months
            - Every season has 楓紅, total 18 months at total
            - Every year, will change salary
        - Has been at TSMC for 7 years
            - Start annual: 1,140,000 NTD
            - Now annual: 2,400,000 NTD
        - If PhD
            - If master’s degree, start at level 31

            - Must promote twice to match the PhD job grade
        - If PhD, start at level 33
        - Level 34 is management level
    - Promotion system
        - If join newer manufacturing sites (like Kaohsiung, Tainan), faster promotion
            - But very tiring, 8am-11pm work days
            - To get a promotion, need the boss to keep seeing you, but very "sell liver"
        - TSMC has two types: fabs and R&D
            - He started as fabs, 設備工程師, then transitioned to process engineering, then finally to R&D
                - Not a very common route to this route, especially to go from factory to R&D
            - For factories, promotions really based on "sell liver"
                - More blood, sweat, and tears
                - Need to put in a lot more time to get promotion
            - For R&D, promotions more based on patents/pattern
                - PND: 考績, decides promotion
                - Must achieve a certain KPI
                - Better work-life balance
                - Do not need to take shifts, regular work hours
        - TSMC really prizes the elitist degrees
            - 北大 graduation, so didn't have many choices when choosing, so start
- Retention rates in the industry
    - NYCU Prof. Becker Hsu
        - Very seldom they leave there, tend to stay for life
        - Very nice package that TSMC and companies offer
            - Good pay, good salary, nice gyms, accommodations, restaurants
        - If they leave the industry, not because they want to change their career, might start up their own business
    - TSMC
        - If at factories, then many will transfer over to 外商 (international companies)
            - Since there is a limit to physical ability of taking shifts, especially when they start having a family
            - Will not offer better salaries, but just offer more freedoms
                - TSMC very prize customer privacy
                - Can't take cellphones or Apple Watches into the office
                - Most websites get blocked
                - BUT more recently, to attract more young talent, have slowly lessened these restrictions (can now take in Apple Watches, can browse some private sites)
        - If at R&D, then more likely to stay all the way until retirement

        - Attracting young talent in recent years
            - Besides aforementioned freedom, loosen some restrictions
            - Salary!! Definitely most of this
                - Increase the start salary of new workers, often to get closer to those of more senior engineers
                - More senior engineers get upset, but TSMC is focusing on the younger talent
        - 挖角: dig out engineers from TSMC
            - Must be very under the table
            - Chinese companies will secretly come to look for managers
            - Personally knew several managers who have been dug up
            - When entering TSMC, sign agreement, cannot within 2 years of leaving to another comparable company (in semiconductor industry)
                - Not strictly enforced, but signed agreement
            - Definitely less within the past couple years: covid, maybe also government
- Government resources provided?
    - NTU Dean Tzi-Dar Chiueh
        - Law is the new experiment of public-private partnership in higher education
        - Requires each new college/graduate school find collaborative company to sign agreement to provide a certain amount of money to university, then government matches that amount
            - 50/50 partnership funding
            - Companies become founders
        - Public universities play a very important role
            - Chiueh salary comes from NTU, but not from the new graduate school
            - NTU university provides third wheel of very important space, classroom, resources
            - New colleges cannot build curriculum in such a short interval
            - Most advisor and faculty come from existing universities
        - 3 different stakeholders in the school
        - Money: 4 companies as founders
            - TSMC, PowerChip, MediaTek, Etron
            - Presidents are all NTU alumni
            - Previously already had very close relationships
            - Approached them for long-term commitment (10 years)
            - Entire graduate school was experimental 10-year project
                - Proposal needed plans for all 10 years, including budget
            - Budget is around NTD 200 million per year
                - $6.5 million per year
                - Half of this comes from industry, other half comes from government
    - NSTC
        - Academic-policy collaboration

- Technology funding
- Important organization in Taiwan to support academic research
- In charge of technology transfer
- Foundation of industry talents
    - Support academic research want to cultivate talent from the university
    - Through university and graduate school training
    - Semiconductor industry development → advantage in the global market
    - Educate more talent
- New policies from the department
    - IC design and IC packaging
- Funding
    - Hard to calculate – not just their department, also civil engineering department
    - Civil engineering department is in charge of funding
        - Focused on chemical materials and semiconductor technology
        - Do not have figures on hand
            - → can give when I visit
            - Have yearbooks about that
        - Some budget from various departments
    - Also natural science department: fund physics research relating to semiconductor
- Collaboration between departments in the NSTC
    - Decide the annual budget before each year
    - A lot of meetings: what technologies they want to develop in the following four years
    - National policy for the following 2-4 years
- Concerns or problems in the next few years?
    - Taiwanese semiconductor industry is the strongest in the world right now
    - Want to maintain status and competitiveness
    - Key issue: keep the status and the talent
    - This type of industry needs a lot of capital
        - A lot from stock market and global financial investment
        - But the talent: is a problem
            - → has already developed very close relationships with Taiwanese universities, but it's not enough
    - New policy: create policies to create new colleges in
        - Also get students from around the world: scholarships, etc.
        - Also production base around the world
    - Big problem: talent
- Executive Yuan's
    - Only care about politics, but not the reality
    - Taiwanese engineers are quite young (30s to 40s), they have families, care about children's education, care most about the environment

        - In the 20 years, some engineers go to mainland China, but they leave their families behind
        - Because of recent
        - Not too worried in recent years whether engineers go over to mainland
        - Big changes in past 5 years:
            - trade war, Trump
            - Covid 19
                - → mainland China suffered heavily because of covid, damaged normal life and production line
    - What if covid 19 is fixed? Changes in next 10-15 years?
        - Hard to imagine
        - But need to keep competency
        - Help industry develop well in the country
            - Focus on talent cultivation
    - In the past, Taiwanese talents good at hardware
        - Now digital transformation, AI, software, smart application
        - Invest more resources in this new type of talent incubation and scientific research
            - Big: smart medicine, digital kiosk, precision medicine
            - Want to develop smart applications, field research application
    - How have company-university partnerships changed over time, particularly in recent years?
        - Most of the budget is from the government
        - But her department: largely from the industry
            - Share results with the government
        - Past few years, noticed a lot of changes in the future (technology advances)
            - Demands from population (ex: carbon emissions as a result of hardware manufacturing, carbon tax)
            - Companies want to reduce carbon emissions in production procedures
            - Digitalization
    - Problem: decreasing population
        - Younger generation not as keen on having children
        - Not just more manpower, but also machines and devices
    - Which companies are most involved?
        - TMSC: four project with NTU, tsinghua, jiaotong, chenggong university
            - NTU: 1 nanometer manufacturing procedure
                - Make smaller and smaller chips
            - Tsinghua: focus on EUV (equipment for manufacturing technology)
            - Jiatong: focus on materials
                - Different materials, mostly currently use silicon, but need new materials that conserve more energy

                    - In the phone, the most energy intensive component is the CPU
                    - When creating new CPU chips, not just smaller and faster, but also more energy efficient
            - China steel
                - Government-owned company
                - Carbon emission issues, green technology
            - 長春: work with Tsinghua for chemical
                - Develop green factory
            - Problems that they can't solve, they leave it to universities to solve
                - And hope that students will go work
        - Each professor is very specialized, so they don't know much
            - And the funding is dispersed
        - How many students go into the industry?
            - Retention rate: hard to tell, especially with military service
                - Also many study abroad
            - Company provides very good salary and bonus
        - Salary and bonus?
            - NSTC has this information, but not off the top of their head
            - Ministry of economic affairs:
                - Has info on salary, background, company, etc.
        - Also has income tax statistics
            - The highest salaries are all in Hsinchu, Zhubei, engineers
        - Scientific survey every year!
            - May also find figures online, and can go
        - How many people migrate to China to work in the semiconductor industry? At what age or point in their career do they choose to do so?
            - → not really, because Taiwan won't ask
            - Anecdotally, not much, most people come back
            - Chinese semiconductor industry aren't doing well
                - And a lot of volatility, not a lot of transparency
        - American trade war has had
        - Most concerned with
        - Executive Yuan probably following American Chip Act
        - People to talk to
            - 前展出 qianzhanchu is the right department for national security
            - Potential problem with confidentiality
- Reasons for recent push
    - NTU Dean Tzi-Dar Chiueh
        - Very large shortage of talent in electronics and semiconductor industries
            - Semiconductor industry is booming on globally
            - Growth rate: 5-10% steady
            - Manufacturing and design for semiconductors have shifted to Asia (Taiwan, Korea, China)

            - Companies grow bigger and bigger
                - TSMC market share growing much faster, very high hiring
            - Demand side is main reason
        - Government passed new law last May:
            - Human talent cultivation and academic-private partnership
            - Public universities set up with private companies
        - Government are more willing to help with semiconductor (national security, public support)
        - Doesn't think this law would have been passed 6-7 years ago
            - Protests from other industries: why invest more in an industry already making so much money
            - National security concerns!!
            - Was watching telecast – was passed in one day, no opposition
    - NYCU Prof. Becker Hsu
        - Not a real shortage, just a relative thing
            - Birth rate is slowing down
            - Semiconductor technology is expanding too fast, expanding territories and new products, so shortage in students
        - Taiwan set up 4 new semiconductor schools in NCKU, NTU, NTHU, NYCU
            - Four main programs: goal is to educate more engineers and students, to devote into expanding technology
- National Security Act amendments will impact the semiconductor industry?
    - NTU Dean Tzi-Dar Chiueh
        - Not too familiar, but believes this to not be as strict as the U.S., where permanent residents can't work for American companies with potential trade secret breaches
        - Strength of Taiwanese passport: allows Taiwanese citizens to work for companies abroad
        - Taiwan will follow the demands of the US. It's a loss of autonomy, but the situation demands it (this prof feels reluctant). US will probably not allow Taiwanese to work for Chinese
            - Generally very split in terms of domestic politics: not just the US pressure, but lots of domestic (⅓ think China is the enemy, another ⅓ think China reunification)
    - NYCU Tom (Becker's student)
        - Knew about this recently, more news coming around, electronic bulletin boards for news on semiconductor
    - NYCU Gwen (Becker's student)
        - Not aware of these at all → may affect answers below
    - NYCU Oliver (Becker's student)
        - Yes, have heard news about this amendment, but don't know many details
            - Doesn't know which ones you can go or not go
    - NYCU Dale (Becker's student)
        - Haven't heard about this before
    - NTUST G2

        - Will impact, since the high-tech managers who understand more will be more restricted
        - If China starts to have more of a semiconductor industry, then there will be more tension
        - Student 2's friend: going to US to study EE graduate school, applied for visa, got asked questions about taking core technologies / trade secrets from US to Taiwan
            - They scoffed a bit at this idea, but emphasized more Chinese visas are rejected
    - TSMC
        - Has heard of this law
- Impact your students' career choices?
    - NTU Dean Tzi-Dar Chiueh
        - 3-5 years ago, Chinese companies offer very good compensation
        - Senior engineers willing to leave home base to go to Shanghai, Guangdong to work for Chinese company
        - But now, more factors:
            - Majority deem semiconductors as national security issue
            - Pressures from friends, relatives
            - "American citizen working for Russian company"--> not just about money
            - Taiwanese engineers very popular abroad, have many places elsewhere to find employment
            - Younger generation more prone to work globally: willing to relocate, and China is one of the foreign countries (Japan, Singapore, US, Australia)
    - NYCU Tom (Becker's student)
        - Hasn't affected him or the people around him
        - TSMC going to US, so maybe new technologies
        - Taiwanese tech not limited to what TSMC is doing
    - NYCU Oliver (Becker's student)
        - Market in China is very huge
        - If have a chance to go work in China, it's not a bad thing; it's a good thing
        - But there are also some national issues between Taiwan and China, so know that there are reasons for this policy to appear
        - But more or less, might impact the chance to work in the big company
        - For friends/other people: China has a big market, so if we lose the chance of working opportunities in China, then lessen opportunities
            - There are also other big companies in the whole world
        - Big companies in China and abroad:
            - 金東方: tech
            - Other semiconductor companies, unsure of names
            - Intel, Samsung, TSMC, european company
- Work for a Chinese company? What were their reasons?
    - NTU Dean Tzi-Dar Chiueh

        - Student did not go to China, but stayed in Taiwan and recruited a team of engineers for a new company
        - Funding from Chinese companies, money from BitCoin, wanted to invest in IC design company
        - Student has worked with IC designs in Taiwan
        - Government feels this may have been a breach of security
            - 兩岸人名關係條例
            - Prohibit Chinese companies to set up shop in taiwan (can't do R&D here)
            - Government feels they're using Taiwanese figurehead for CHinese company to do IC research in Taiwan
            - Many taiwan engineers would rather not work in China but work in Hsinchu
        - All over the news!
            - Student wasn't charged, many prosecutors
            - Police searched company → very big deterrent gesture by government
        - Chinese companies set up branches in Hsinchu, get engineers
            - Believe this practice is no longer allowed
        - Still people go over to Shanghai
            - Doesn't prohibit Taiwanese going over to work for Chinese companies in IC design
    - NYCU Prof. Becker Hsu
        - Semiconductor technology is blooming in Taiwan
        - All the engineers in Taiwan in this industry are very well paid, so very rarely do they leave to work abroad, especially in the semiconductor industry
            - Do not know any lab graduates going to leave Taiwan
    - NYCU Tom (Becker's student)
        - Not really, working for Taiwanese company is much easier
        - Don't like the environment in Beijing, was in Beijing before, "not in my interest"
        - Recent culture is very different from Taiwan, even in their science park
            - Was in Beijing during undergraduate study
            - "More stressed" "it's already stressed here"
            - They work for similar hours, but the breaks are quite short, even weekends on duty
        - For people who do: he doesn't know much about 挖角, but it's in the news often. He understands it's just money, others may see it as more political
    - NYCU Gwen (Becker's student)
        - If they offer a great salary, then she'll go. Why not? Never been there before, want to experience culture
    - NYCU Oliver (Becker's student)
        - Haven't thought much about this part, but yeah?
        - If someone offers more money, then consider what the company does and combination development ability (中和發展性) (wrong zhonghe, essentially ability to work with others to develop in many fields)

        - Ex: if that company has many collaborations with other companies, then the company may not be easily substitutable. Different fields exposed and opportunities
            - But if company close the door and do things by themselves, but may lose to other companies that do collaborate with each other
    - Salary is not the most important, instead the other types of 福利 (benefits; company can take care of a lot of things, like education of children, health insurance, etc); cover more aspects of life; does not want to just go to a company that offers a lot of money and expects him to work work work
- NYCU Dale (Becker's student)
    - No, never
    - Friends and relatives told him that experience with Chinese companies will lead to confidential problems
    - He will never go to China
        - If you earn RMB, then you cannot transfer all that to a Taiwanese bank account
        - Political issues: not sure about the life there
    - Very few people decide to go over
        - Technology and things you actually do there – they are limited
        - Issue of equipment:
            - 紫外光機
                - → very very important!
                - They use equipment from 10 years ago, so bad yield, higher costs (more steps than newer equipment)
                - Low yield, high cost → bad for chip market
            - Critical equipment for semiconductors are forbidden to be sold to Chinese companies
            - Equipment from ASML, from the Netherlands
        - Political issues
        - Taiwan has many great companies in this field, so to stay here, you have more chances to get better to go to another company (if TSMC first, then Mediatek, Qualcomm)
- NTUST G2-1
    - Decline to answer
    - → this student warned the other one to not speak too much, since their professor is particularly sensitive about these topics
        - Prof is even sensitive to buying Samsung devices, because of the competitiveness in the chip industry
    - Asked multiple times to make sure we keep confidentiality
    - It's still sensitive politics at the end of the day
- NTUST G2-2
    - Will, if they offer enough of a salary
    - 10x the salary of Taiwan

                - Not 5x though, because places like TSMC, with EDA (electrical design automation) tools, more advanced software technologies are from the US
                    - The US provides these tools to Taiwan, but not to China
                    - Cadence, Synopsis: 2 of the biggest (only) companies that make these software
                        - Can do less than 3mm
                        - US stops the exports, restrictions
                    - China: EDAs still only limited to 5mm
                - In terms of economy, politics
    - TSMC
        - No → because value democracy
        - Chinese companies offer 5x-6x
        - If yes, what would they have to offer to surpass that of a Taiwanese company?
        - If no, what are the reasons? Would there be a financial offer they could make to counteract these reasons?
- **Financial offer to work at a Chinese company**
    - NYCU Tom (Becker's student)
        - One of his friends got information, looking for job and received offers
        - Salary is 2x-3x compared to offers in Taiwan, can be attractive for many people, especially short period money, "in short period can get much paid"
            - Level: starting engineer
            - Senior engineer in taiwan doesn't even make this much money
    - NYCU Gwen (Becker's student)
        - China: 1.5x salary
        - US: 2x salary, since basic expenditure much higher than Taiwan's
        - Factors: culture and salary
        -
        -
- What are the aggregate statistics of the exit careers of each graduating class? How have these statistics changed over time?
    - NTU Dean Tzi-Dar Chiueh
        - First academic year for graduate school, so no statistics yet
        - His own statistics for Master's and PhD students that have graduated
        - Very large portion go to work for local IC design companies as first job
        - Afterwards, gradually migrate to other positions
            - B&Q, VP of IT (like "Home Depot", retail)
            - Students in counseling positions: very stressed engineers in Hsinchu Science Park, desire to go into new line of work
            - Many still work in IC design companies, most still in electronic industry in Taiwan
            - 20-30% then move on to other degrees and jobs in US
                - Write recommendation letters
            - 20% find employment in US
            - After graduating BS in Taiwan, then graduate

- Political hesitancy to speak on this topic
    - NTU Dean Tzi-Dar Chiueh
        - Not all people are comfortable talking about relationship with China
            - Quite dynamic
            - People used to be happy taking 5x salary and going to China
            - But now, it's less (maybe 50% more) and risk reputation (among friends, spouse, etc)
        - 否極泰來
        - Swinging towards more hostility
            - China will not take this lying down → stop thinking what the relation will be, no matter what restrictions US put on China, there's no response, they just take those hits
        - Careful and prudent about how they will react
        - Students just want to graduate, get a good job, get a good salary
            - If longer term: think about going abroad
            - Need to interview the students
        - Professor happy to speak their minds
        - Engineers do not have time, and may be reprimanded for speaking, priorities are with job
            - Very foreign idea to talk to researchers doing survey
            - HR doesn't know exact current technical status
            - First line is the engineer knowing the technical difference between China
        - People working for TSMC don't know what TSMC is going to announce
            - Very compartmentalized between departments
            - Don't share
        - Very not willing to talk about retention rate
            - Not researchers or reporters
        - Can get second hand, CANNOT get first hand
        - Ask 畢業生輔導室, career placement office
            - Job fair, big bosses from banks and etc.